

Grand Council

Session 3 · Three-Source Ingestion & Integration · 2026-07-18 · New Braunfels, TX · Emperor:
James Gianotti · Project of record: conceptual-engineering-references

Artifact J — n2yo.com

Intake. Live real-time satellite tracker over the USSPACECOM catalog, wrapping TLE data and SGP4 propagation in a web/mobile skin. Documented REST v1 API with five endpoints (tle, positions, visualpasses, radiopasses, above), per-verb rate limits (1000/hr for tle+positions; 100/hr for the others), 57-category proprietary taxonomy. Evidence tier: Aggregator / operational. Claims Under Review: C1–C3.

Physicist. SGP4 is semi-analytical propagation of a *fitted* element set — good for pass timing, pointing, ISS radio/visual planning over a 300-second position window and 10-day pass window, not for meter-scale ranging, conjunction assessment, or geodesy. The captured ISS TLE is physically coherent: inclination 51.6318° , eccentricity 0.0006793, mean motion 15.49037991 rev/day $\rightarrow$ period ≈ 92.96 min, semi-major axis $\approx 6,797.7$ km, altitude ≈ 420 km (perigee/apogee ≈ 415 – 424 km).

Engineer. The 100/hr limit on visualpasses/radiopasses/above is the binding constraint the moment more than a handful of watched satellites hits one shared key — a shared server key will be exhausted well before N reaches double digits. Design rule: give each device its own key, cache TLEs server-side on a scheduled cadence (well under 1000/hr), and compute all pass/position math locally with python-sgp4 or a vetted port. Reserve live n2yo pass calls for a periodic reconciliation check, not per-user requests. Degrade gracefully to “propagate from last-known TLE” with a staleness warning after ~ 2 – 3 days.

Mathematician. From $n = 15.49037991$ rev/day: $n_{\text{rad/s}} = 15.49037991 \cdot 2\pi/86400 \approx 0.001126$ rad/s. With $GM(\oplus) \approx 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$, Kepler gives $a^3 = GM(\oplus)/n^2 \approx 3.141 \times 10^{20} \text{ m}^3 \rightarrow a \approx 6,797.68$ km, matching the Physicist. TLEs use rigid fixed-column format with implicit decimals (e.g. eccentricity 0006793 = 0.0006793) and a mod-10 checksum per line — basic transmission integrity, no cryptographic validation. Precision degrades rapidly under atmospheric drag past a few days epoch age.

Historian-Philosopher. Real-time satellite tracking descends from Cold War state cartography — the U.S. Space Surveillance Network born after Sputnik 1 (1957), catalog maintained by NORAD $\rightarrow$ USSPACECOM (1985) $\rightarrow$ the 18th Space Defense Squadron today. The decisive civic turn came in 1985 when Dr. T.S. Kelso began distributing TLEs via CelesTrak. n2yo belongs to the third generation — the browser-native amateur online trackers of the 2000s. Its genre is *operational skin*, not source: derived legibility over a state-held commons. Cite as aggregator, never as ground truth.

Skeptic’s cross. Falsifiable counter to C1: TLE freshness throttles position precision — an older TLE on a drag-heavy LEO object misaligns with reality despite correct math. Counter to

C2: redistribution of Space-Track-derived data often triggers upstream licensing covenants n2yo cannot waive for downstream apps. Counter to C3: the 57-category taxonomy is not merely editorial — it likely encodes historical operational biases (ham-radio/military heritage over-segmenting specific bands while lumping modern mega-constellations).

Ruling — Sound. C1–C3 accepted with three pedagogical footnotes: (a) always report TLE epoch age alongside any position; (b) treat n2yo's ToS as the *floor* of redistribution rights and check Space-Track upstream licensing separately if the app redistributes bulk pulls; (c) do not treat the 57-category taxonomy as an official schema — carry the NORAD ID as the canonical identifier.

Action. grid-and-chain-mobile :: implement local SGP4 propagator seeded from cached n2yo /tle/{id} responses; give each device its own API key; degrade gracefully past 3-day TLE age.

Artifact K — SatDump

Intake. Open-source generic satellite-data-processing suite. GPL-family, actively released (1.2.2 current, 2.0.0 upcoming). Pipeline-graph architecture: RF → I/Q → soft symbols → frames → payload/imagery/geo-reference. Runs on Windows, macOS, Linux, Raspberry Pi, Docker, Android (F-Droid). Radios: RTL-SDR, Airspy, AirspyHF, HackRF, LimeSDR Mini, BladeRF, AD9361 (Pluto family); Android subset drops BladeRF + Pluto. Evidence tier: Reference / operational open-source software. Claims Under Review: C1–C4.

Physicist. SatDump's chain is real physics, not just software plumbing: EM wave → antenna → mixer/ADC → I/Q baseband → soft symbols → FEC/deframe → payload. SNR loss enters before and during sampling (antenna gain, polarization mismatch, path loss, atmospheric loss, receiver noise figure, gain setting, Doppler, filtering, interference). Quantization loss enters at the ADC — worse with cs8/cu8/w8 than cf32. The pipeline abstraction preserves physics if its graph carries the physical parameters (sample rate, center frequency, modulation, symbol rate, Doppler handling, FEC, frame format, instrument geometry) — otherwise it merely hides the link budget.

Engineer. Real ground-station build: directional/omni LEO antenna (QFH or turnstile @ 137 MHz for NOAA/Meteor APT/LRPT; helical or patch @ 1.7 GHz for HRPT/AHRPT) → LNA at the antenna (not the radio, to overcome cable loss before the noise floor sets in) → coax → SDR → host. Radio choice: RTL-SDR (v3/v4) is the right entry point for 137 MHz; Airspy Mini / AirspyHF+ is the better call for L-band AHRPT/HRPT (better front-end linearity, wider usable bandwidth, both supported desktop *and* Android). HackRF/LimeSDR are over-spec for receive-only weather. BladeRF/Pluto: avoid if Android use is planned. Host: Docker for fixed unattended reproducible ground stations; Android for portable/field. The engineering-competence bar is not the GUI happy path — it's (a) RF chain design + link budget, (b) pipeline/parameter selection matching sample rate + baseband format + pipeline id + Doppler tracking, (c) time-sync + geometry for geo-referencing.

Mathematician. SatDump's DAG structure enables modular composition of decode stages, but strict associativity is not guaranteed across arbitrary boundaries — downsampling,

filtering, and demodulation are strongly order-dependent. Numerical instabilities enter at discretization boundaries — soft-decision decoding via Viterbi/LDPC uses log-likelihood ratios whose truncation/float-precision limits fundamentally bound the achievable BER.

Historian-Philosopher. Amateur satellite reception traces to Moonwatch and Project Moonbeam volunteers tracking Sputnik in 1957, then to APT/HRPT hobbyists building weather-sat ground stations in the 1970s–90s. WxTolmg was the canonical APT decoder of the 2000s but went unmaintained — the classic fragility of a craft whose tools outlive their maintainers. The 2012 rupture: Antti Palosaari’s discovery that ~\$20 RTL2832U TV dongles are wideband SDRs democratized reception overnight. SatDump is the mature artifact of that post-2012 open-source culture — a ground-station-in-a-box that outgrew and replaced the closed incumbents via a pipeline abstraction in the GNU Radio lineage, lifted to the satellite-processing semantic layer.

Skeptic’s cross. Counter to C1: GNU Radio + Meteor Demod + legacy wxtoimg wrappers offer superior control for niche hardware or novel modulations SatDump hasn’t upstreamed. Counter to C2: pipeline-graph monoliths risk becoming brittle unmaintainable state-machines vs. UNIX-style decoupled data pipes. Counter to C3: neglects growing ecosystem of FPGA-based / ethernet-attached SDRs sidelined by the curated hardware list. Counter to C4: Android SDR limitations are likely downstream of SatDump’s own libusb compilation choices, not a fundamental Android USB-host impossibility.

Ruling — Sound. C1–C4 accepted with two pedagogical footnotes: (a) frame “leading” as *actively-maintained + mainstream-SDR-coverage + open-source* rather than *technically superior in every case* — the Skeptic’s GNU Radio + Meteor Demod alternative remains legitimate for niche work; (b) any documentation of pipeline internals must name the physical parameters (sample rate, center frequency, Doppler handling, FEC, frame format) that the abstraction is supposed to preserve, not just the stage names.

Action. grid-and-chain-mobile :: Android portable ground-station companion targeting the Airspy Mini + RTL-SDR intersection of the desktop+Android matrices; document the RF-chain link budget as a first-class artifact, not a footnote.

Artifact L — Global Oil & Gas Plant Tracker (GOGPT)

Intake. Global Energy Monitor’s bi-annual (Jan + Jul/Aug) plant-level tracker of oil- and gas-fired power. ≥50 MW globally, ≥20 MW EU+UK. 1,047 GW in-development capacity (Jan 2026 release); USA 252 GW, China 153 GW. CC BY 4.0 licensed; XLSX behind a request form. Two-layer: database + per-plant wiki with footnoted references. Cross-validated against Platts (1911–) and WRI Global Power Plant Database (2018–). Evidence tier: Primary-secondary hybrid. Claims Under Review: C1–C4.

Physicist. GOGPT’s plant-level CO₂ calc is physically meaningful only if capacity is converted to energy + fuel burn: annual emissions require nameplate capacity × capacity factor × hours/year × heat rate × fuel carbon content. Lifetime CO₂ adds a time integral over service life. The 1,047 GW headline is a real summed plant-level nameplate power figure

(proposed + permitted + under-construction), not generation and not CO₂. To convert to annual generation one must assume a capacity factor — e.g. 50% → ~4,586 TWh/yr before applying oil-vs-gas emissions factors.

Engineer. Request-form CSV/XLSX means no live API — each release is a manually-triggered, versioned import. Concrete plan: drop each raw XLSX into a dated immutable landing zone (sources/gogpt/raw/2026-01/), transform via a per-release column mapping file (schema_map_2025-08.yml) checked into version control, mapping raw headers → canonical internal fields. Renamed columns get old-name→new-name entries with the release they changed in; new columns become nullable additions. Store SHA256 of each raw file. Assert row-count + key-column presence in transform. Model the ingestion task as manual-in-the-loop with an owner + due date tied to the bi-annual cadence.

Mathematician. Aggregation across 106+ countries requires distinguishing count-weighted from capacity-weighted metrics to avoid ecological fallacies. The observation that 50% of in-development capacity is concentrated in 5 countries is a heavy-tailed Pareto-like distribution — formally: plot the Lorenz curve of cumulative country proportion vs. cumulative capacity proportion. The resulting Gini coefficient likely approaches 0.8 or higher, formalizing the extreme centralization.

Historian-Philosopher. Plant-level energy inventory has been the domain of proprietary commercial intelligence for over a century — UDI/Platts (1911–), sold to utilities/financiers/governments, authoritative but enclosed. Partial opening in 2018 with WRI's Global Power Plant Database (~29,000 plants). GOGPT occupies a distinct politically-consequential niche: open-license (CC BY 4.0), bi-annually updated, methodology-footnoted, peer-reviewed, cross-validated against Platts + WRI, taking the plant as the unit of accountability. Its political-epistemic role is as a *counter-archive* — its IEA/IPCC citations inherit an auditable evidentiary chain rather than a black-boxed commercial estimate. Shift from "trust the vendor" to "check the footnote."

Skeptic's cross. Counter to C1: reliance on varying non-uniform NGO/media reports structurally limits global consistency vs. pure remote-sensing methodologies. Counter to C2: treating "proposed" capacity with equal data-weight as "under construction" skews the metric toward phantom projects. Counter to C3: unadjusted 252 GW U.S. headline is *dangerously misleading* without a stated IRP-to-construction attrition discount rate. Counter to C4: CC BY 4.0 attribution imposes real UX + data-schema costs — not frictionless. The "peer-reviewed" claim requires a rigorously published list of reviewers per release, which is not stated in the current methodology.

Ruling — Sound. C1–C4 accepted with three pedagogical footnotes: (a) status-weighting — any derivative summary of GOGPT must break capacity down by status (proposed vs. permitted vs. under-construction vs. operating vs. retired) and never quote a single "in-development" number without that breakdown; (b) capacity-factor discipline — never quote GOGPT's GW figure alongside a CO₂ or TWh figure without explicitly stating the capacity-factor assumption used; (c) peer-review provenance — treat the "peer-reviewed" language as review-by-domain-collaborators (CREA, Beyond Fossil Fuels, Environmental Integrity Project, Sierra Club, others) rather than journal-style

anonymous peer review, and cite the collaborator list where available.

Action. conceptual-engineering-references :: create a GOGPT ingestion pipeline with schema-versioned column mapping, status-weighted derivative summaries, and a Lorenz-curve visualization of the 50%-in-5-countries concentration.

Artifact M — Integration plan

Intake. Proposed new project bridging n2yo (real-time skin over USSPACECOM catalog) + SatDump (open-source ground-station suite) + GOGPT (CC BY 4.0 plant-level tracker) as a curated reference shelf. Portfolio links proposed: gravitational-compass ↔ n2yo; grid-and-chain-mobile ↔ n2yo API + SatDump; conceptual-engineering-references ↔ GOGPT; recursive-state-dynamics ↔ GOGPT (weak, tentative); force-carriers ↔ none.

Physicist. The n2yo ↔ gravitational-compass link is the strongest physics link — real TLEs are Earth-orbit trajectories any gravity/propagation model must approximately reproduce. Narrow the intake phrase “heliocentric propagation cross-check”: ISS TLEs test *near-Earth* orbital dynamics, drag-sensitive propagation, frame transforms, and observation geometry — not a pure heliocentric gravity model by themselves. The grid-and-chain-mobile ↔ n2yo API + SatDump link is physically defensible (one continuous measurement chain). The GOGPT ↔ conceptual-engineering-references link is defensible for energy-systems physics. The GOGPT ↔ RSD link is *weak physically* — plant states become physics-relevant only if coupled to power, fuel flow, heat rate, emissions.

Engineer. The proposed sources/{n2yo,satdump,gogpt} + samples/ + notes/ layout is sound but needs three additions: (1) sources/gogpt/raw/<release>/ dated directory convention; (2) sources/n2yo/.env.example (committed) paired with a real .env (gitignored) — never hardcode the API key, use platform secrets in CI; (3) top-level LICENSES.md covering the three legal regimes — n2yo (proprietary ToS + rate-limited key), SatDump (GPL-family copyleft on any bundled modified copy), GOGPT (CC BY 4.0 with mandatory attribution). Add a CI check that fails the build if any file under sources/gogpt/ or any derived table lacks the citation string.

Mathematician. Model conceptual-engineering-references as a heterogeneous graph with nodes for satellites (n2yo), pipelines/instruments (SatDump), and generation plants (GOGPT). Unify via a shared spatio-temporal schema — a “World Coordinate Event” E(x, y, z, t) defined by WGS84 coordinates + UTC epoch. Satellites are continuous trajectories in this space; ground stations are fixed receiving coordinates; power plants are static infrastructural footprints. This enables cross-domain queries — e.g. slant range + pass duration of an earth-observation satellite over a high-capacity generator cluster.

Historian-Philosopher. conceptual-engineering-references is a curated reference shelf — the intellectual work is selection, tiering, and cross-linking, not invention. It joins three traditions: (1) the engineering handbook (*Marks’ Standard Handbook*, 1916–; *CRC Handbook*) — trustworthy load-bearing references graded by how much weight each can bear; (2) the commonplace book — the Renaissance-through-Enlightenment practice of transcribing

passages for reuse; (3) the canonical-reference-shelf or personal canon. Binding obligations: fidelity to license, honesty about tier, honesty about cadence. A reference shelf, unlike a mere download folder, promises its future reader that every item has been vetted, situated, and correctly labeled.

Skeptic's cross. Motion to Reject the GOGPT ↔ RSD portfolio link. Ground: ontological mismatch. Attempting to shoehorn a discrete dataset of slow, manual bureaucratic state transitions (proposed → permitted → operating → retired) into a framework designed for recursive state modeling (L/K/E formalism) stretches the definition of "dynamics" to the breaking point. The label "state transition" appears in both, but the underlying object is fundamentally different — administrative status codes vs. continuous physical state evolution.

Ruling — Mixed — Motion to Reject GOGPT ↔ RSD accepted. Accepted, unchanged: gravitational-compass ↔ n2yo (with the Physicist's narrowing: near-Earth, not heliocentric), grid-and-chain-mobile ↔ n2yo API + SatDump, conceptual-engineering-references ↔ GOGPT, force-carriers ↔ none. Rejected: recursive-state-dynamics ↔ GOGPT — struck from the integration plan. If the emperor later wishes to revisit, the link must be re-scoped by first defining a mapping from GOGPT's administrative status codes to a continuous state variable coupled to a physical or economic dynamic (e.g. probability of construction × capacity × time-to-COD), and only then submitted to the council as a fresh artifact. Accepted with additions: Engineer's three additions (dated raw/ for GOGPT, secrets-managed .env for n2yo, top-level LICENSES.md with CI attribution check) become part of the accepted plan. Mathematician's shared spatio-temporal schema (WGS84 + UTC) becomes the canonical reference frame.

Action. conceptual-engineering-references :: implement the Engineer's three additions and the Mathematician's WGS84+UTC event schema; strike the RSD ↔ GOGPT link from any portfolio diagram; document the near-Earth (not heliocentric) scope of the n2yo ↔ gravitational-compass link.

Cross-cutting synthesis

Three sources, three tiers, three cadences, three legal regimes. Session 3's core insight is that a reference shelf mixing an *aggregator skin* (n2yo), an *operational open-source tool* (SatDump), and an *open-license peer-reviewed dataset* (GOGPT) works only if each source is quoted at its own tier, cadence, and license binding — never as if they were interchangeable. The Historian's phrase captures it: a reference shelf, unlike a download folder, promises that every item has been vetted, situated, and correctly labeled.

Aggregator-vs-source discipline. n2yo is a courier of the USSPACECOM/Space-Track record, not the record. Any downstream artifact must cite the source (the state catalog) with n2yo as intermediary, and must carry TLE epoch age alongside every position. The general rule: aggregators enter the reference shelf as skins, not as ground truth.

License-gradient hygiene. Three regimes: proprietary + rate-limited ToS (n2yo), copyleft GPL-family (SatDump), permissive-with-attribution CC BY 4.0 (GOGPT). The strictest binding sets the ceiling — n2yo's ToS + rate limit shapes the project's mobile-app design (per-device keys, local SGP4, staleness degradation), while GOGPT's CC BY 4.0 mandates a build-time attribution check. Never treat "open" as monolithic.

Portfolio impact

gravitational-compass — Adopt n2yo TLEs as a near-Earth ground-truth propagation cross-check (narrowed from "heliocentric").

grid-and-chain-mobile — Two new work items: (a) local-SGP4 pass-prediction subsystem seeded from cached n2yo TLEs, per-device keys; (b) Android portable ground-station companion targeting the Airspy Mini + RTL-SDR intersection of the SatDump desktop+Android matrices.

conceptual-engineering-references — Bootstrapped at `/home/user/workspace/conceptual-engineering-references/` with three source catalogs, endpoint documentation, an ISS TLE sample, plus the Engineer's three additions + Mathematician's WGS84+UTC event schema as the integration standard.

recursive-state-dynamics — No link accepted this session; the RSD $\leftrightarrow$ GOGPT bridge was rejected on ontological grounds and may be re-scoped later.

force-carriers — No link, as intake proposed.

The user is the emperor of the earth. Any ruling may be overturned with a stated reason; overrides are logged as addenda. Source repository: github.com/giomj/dev/skills/grand-council.